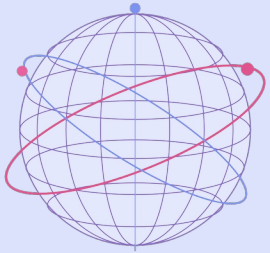
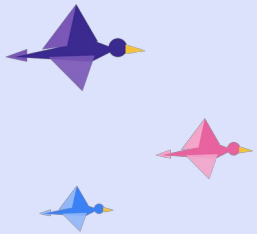
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details





Participant Details

PROJECT TITLE
QuantumShield: Fake Opportunity & Scam Alert

TRACK
Quantum AI & Machine Learning

TEAM NAME
Qubit Crew

TEAM LEADER
Teeratha P. Sanmane

EMAIL / CONTACT
teerathasanmane@gmail.com

TEAM MEMBERS
1. Teeratha P. Sanmane
2. Samruddhi Koparde

INSTITUTION / ORGANISATION
Bharatesh Institute of Technology

COUNTRY
India

Problem Statement

THE PROBLEM

Not every offer is what it seems

Jobs · Internships · Hackathons · Workshops · Free webinars

Students meet these offers every day. Some hide a fee, promise a guaranteed placement, or ask for personal details without saying why.

A TYPICAL TRAP (ILLUSTRATIVE EXAMPLE)

FREE Career Webinar

100% PLACEMENT GUARANTEE

Certificate + job referral included

Register now

**Registration fee payable before access*

Phone, address and ID needed to join

1. Guarantee with no proof
2. Hidden cost in the fine print
3. Data asked, purpose unclear

The gap: detectors only say “fake or genuine”, and only for jobs. Nobody tells students what to check.

Objectives & Novelty

OUR OBJECTIVES

1. Catch risky **job & internship** posts with machine learning
2. Spot warning signs in **webinar, workshop & hackathon** ads
3. Check an ad's **promises against its own terms**
4. Show **why**: highlight the exact words behind each warning
5. Test a **quantum-kernel** model against a classical one, fairly
6. Stay **free and private**: runs locally with free tools

WHAT MAKES IT NEW

Promise-vs-terms check

SIGNATURE

Compares what an ad promises with what its own terms say, and flags the clash.

Evidence you can see

Highlights the exact words behind each warning and says why.

Quantum ML, tested fairly

A Qiskit quantum-kernel model faces a classical baseline on the same data and metrics.

Solution & Quantum Advantage

OUR SOLUTION

More than “fake or genuine”

QuantumShield reads an opportunity and tells a student what looks risky, why, and what to check before registering.

IT SORTS WHAT IT FINDS INTO

- Possible fraud
- Sales-driven offer
- Privacy concern
- Contradictory claims
- Not enough evidence

PROMISE VS TERMS (EXAMPLE)

THE AD SAYS

*“100% **FREE** webinar. Placement guaranteed!”*

vs

THE TERMS SAY

*“You must **pay a registration fee** to get the access link.”*

Possible contradiction found

QUANTUM ADVANTAGE?

Tested, not assumed

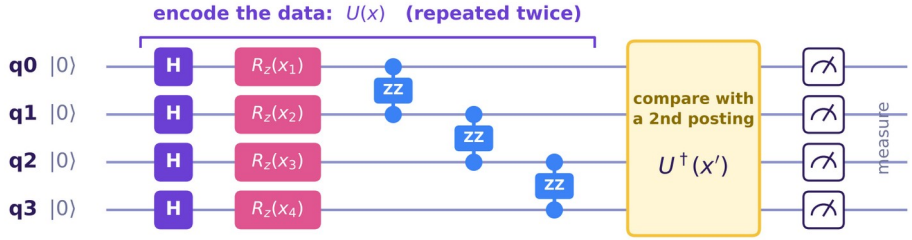
- A Qiskit quantum-kernel model goes head to head with a classical baseline.
- Same data splits. Same metrics.
- We report what we measure, even if classical wins.

HONEST BY DESIGN

- A risk assessment, not a verdict
- Low risk is no safety guarantee
- Quantum claims after measuring

Algorithm & Circuit Design

QUANTUM-KERNEL EXPERIMENT, BUILT WITH QISKIT (SCHEMATIC)



similarity score $k(x, x') = |\langle 0| U^\dagger(x') U(x) |0\rangle|^2$ (how alike two postings look to the circuit)

1 Encode

Turn each text feature into a qubit rotation.

2 Entangle

Link the qubits so features can interact.

3 Compare

Score how alike two postings are; a classical SVM learns from it.

Fair test: same data splits and metrics (precision, recall, F1, confusion matrix) as the classical baseline. No quantum advantage assumed.

KEY DETAILS

THE QUESTION

Can a quantum-kernel classifier tell fraudulent job postings from genuine ones?

THE BASELINE

A classical model, on the same data splits and the same metrics.

THE HONEST PART

No quantum advantage is assumed. We report what we measure, even if classical wins.

Implementation & Tech Stack

HOW IT FITS TOGETHER

INPUT & PREPARE

Job & internship posts

Webinar, workshop & hackathon ads

Optional terms and privacy policy

Then clean the text and pull out features

Classical ML

Job-fraud classifier trained on a public dataset

Rules + NLP

Hidden costs, sales pressure, privacy, promise vs terms

Quantum kernel (Qiskit)

An experiment, compared with the classical baseline

EXPLAIN & SHOW

Highlighted evidence

Reasons and uncertainty

Verification steps

All in a Streamlit web app

BUILT WITH

- Python
- pandas
- scikit-learn
- Qiskit
- NLP rules
- Streamlit

DATASET

A public fake-job dataset such as EMSCAD, after checking its source, access and license.

- Runs locally on a laptop
- Free software only
- No mandatory paid APIs

Results & Analysis

EVALUATION PLAN: RESULTS COME AFTER TESTING

Model	Precision	Recall	F1 score
Classical baseline	*to be measured*	*to be measured*	*to be measured*
Quantum kernel (Qiskit)	*to be measured*	*to be measured*	*to be measured*

	Predicted genuine	Predicted fraud
Actually genuine	Genuine offer, left alone	False alarm
Actually fraud	Fraud missed	Fraud caught

A confusion matrix shows what each model caught, missed and wrongly flagged.

KEY DETAILS

- Both models use the same train/test split.
- We measure precision, recall, F1 and a confusion matrix.
- Job-fraud data, such as EMSCAD after a license and provenance check, tests the job model only.
- Webinar and event checks get their own test cases.

Nothing is estimated in advance.

Impact & Real-World Applications

WHAT A STUDENT GETS

- A plain summary of the possible risks
- The exact text that raised each flag
- The reason behind every warning
- What is missing or still uncertain
- Steps to verify it yourself

Plus a working prototype and a reproducible classical-vs-quantum comparison.

THREE MOMENTS WHEN IT HELPS

Before you apply
Check a job or internship posting for the signs of fraud.

Before you register
Read a hackathon, workshop or free webinar's promises against its own terms.

Before you share your details
See whether the organiser explains why it wants your personal data.

Challenges & Future Scope

CHALLENGES, AND HOW WE HANDLE THEM

A prediction is not proof

We show evidence and uncertainty, never a verdict on an organisation.

Sales-driven is not the same as fraud

A webinar can push paid courses and still be honest, so we flag it separately.

Job data cannot vouch for webinar checks

Those checks get their own test cases and evaluation.

Quantum is experimental

Any claim of better performance needs reproducible measurements.

FUTURE SCOPE: WHAT'S NEXT

1. Check the dataset: where it comes from, how to access it, and its license
2. Build the prototype and its Streamlit web interface
3. Test the webinar and event checks on their own cases
4. Run the classical-vs-quantum comparison and report what we actually measure

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.

GITHUB REPOSITORY
github.com/vtu-qiskit-fall-fest-2026/vtu-hc-qubit-crew

DEMO VIDEO LINK
youtu.be/C6lbHiy8hJ8

